

MICROPROCESSOR & MICROCONTROLLER

A PROJECT REPORT

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THE MAHARAJA SAYAJIRAO UNIVERSITY of BARODA

Faculty of Technology & Engineering

Electronics Engineering

2025-2026

CERTIFICATE

DATE: October 2025

This is to Certify that, Varma Swapnil N., Sarvaiya Snehal., PRN: - 8024058406, 8021080217., of Bachelor's in Electronics Engineering have successfully completed the **Term-Work of Microprocessor & Microcontroller “Interface 4-digit 7 seg. LED display which receives serial data from PC and display the received character as well as three following characters. E.g., if character '1' is received then display 1234. if character 'B' is received then display BCDE.”** offered during the academic year 2025-2026.

Sign. of
Project Guide

Sign. of
Head of Dept.
Electrical Engg. Dept.

ACKNOWLEDGEMENT

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ABSTRACT

This project presents the design and implementation of an embedded system that interfaces a 4-digit seven-segment LED display with an 8051 microcontroller to display serial data received from a personal computer (PC). The primary objective of the system is to receive a single character via the serial port and automatically display that character along with the next three consecutive characters in sequence. For example, when the character '1' is received, the display shows "1234," and when 'B' is received, the output is "BCDE."

The system is developed using the 8051 microcontrollers, which manages both the serial communication and the multiplexing of the 4-digit seven-segment display. Serial communication is established between the PC and the microcontroller through UART protocol at a predefined baud rate. Upon receiving a character, the microcontroller converts it into its ASCII value, increments the value to determine the following three characters, and maps them into their respective seven-segment codes for display. Multiplexing is used to drive all four digits efficiently, ensuring minimal hardware requirements while maintaining clear and stable display output.

This project highlights the practical integration of serial communication and display interfacing, which are essential concepts in embedded systems. It demonstrates how microcontrollers can be programmed to handle real-time data acquisition, conversion, and display operations. The system can be extended to applications such as alphanumeric displays, communication devices, and real-time information systems. Overall, the project provides a strong foundation in serial communication, display multiplexing, and microcontroller programming, making it an effective academic exercise as well as a stepping stone for more advanced embedded applications.

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CHAPTER 1 INTRODUCTION

1.1 Background

In today's digital era, microcontrollers have become the backbone of many electronic systems, ranging from consumer appliances to industrial automation. Among the fundamental applications of microcontrollers is the ability to communicate with external devices and to display information in a human-readable format. One of the simplest yet widely used display devices is the seven-segment display, which can be used to represent numeric as well as limited alphanumeric characters.

The advancement in serial communication protocols has enabled efficient data transfer between computers and microcontrollers. The Universal Asynchronous Receiver Transmitter (UART) protocol is among the most commonly used methods for serial communication because of its simplicity, reliability, and widespread support. This project utilizes UART to establish a communication link between a personal computer (PC) and an 8051 microcontroller. The received serial data is then processed and displayed on a 4-digit seven-segment LED display.

This project not only demonstrates interfacing and multiplexing techniques but also highlights the real-time processing ability of microcontrollers to handle both data communication and display control simultaneously.

1.2 Problem Definition

The problem addressed in this project is to design a system capable of receiving a single character input from a PC and displaying that character along with the next three consecutive characters on a 4-digit seven-segment display. For instance, when the input character is '1', the output should be "1234." Similarly, if the input is 'B', the output should be "BCDE."

This requires the system to:

1. Receive a serial character from the PC.
2. Process the character to identify its ASCII code.
3. Generate the next three consecutive characters.
4. Convert each character into its corresponding seven-segment code.

5. Display the sequence across the four digits using multiplexing.

1.3 Objectives

The main objectives of this project are as follows:

1. To design and implement an interface between an 8051 microcontroller and a 4-digit seven-segment LED display.
2. To establish / between a PC and the microcontroller using the UART protocol.
3. To receive, process, and display characters on the seven-segment display in real time.
4. To demonstrate the concept of display multiplexing for efficient utilization of hardware resources.
5. To provide a practical understanding of combining communication and display systems in embedded applications.

1.4 Methodology Overview

The methodology adopted for this project involves both hardware and software implementation:

1. **Hardware:** The system uses an 8051-microcontroller connected to a 4-digit seven-segment display. The display is driven through multiplexing, which reduces the number of required input pins while ensuring smooth visual output. A MAX232 IC or USB-to-Serial converter may be used for establishing a reliable communication link between the PC and the microcontroller.
2. **Software:** The microcontroller is programmed using embedded C (Keil IDE) to handle UART communication and display control. The program is designed to read the incoming serial character, generate the subsequent three characters, map them into seven-segment codes, and update the display using multiplexing techniques.

1.5 Scope of the Project

The scope of this project extends to educational and practical applications, including:

1. Learning and understanding UART-based serial communication.
2. Gaining hands-on experience with seven-segment display interfacing and multiplexing.
3. Exploring ASCII character handling and conversion in embedded systems.
4. Building a foundation for advanced applications such as alphanumeric displays, scrolling messages, and communication-based embedded systems.

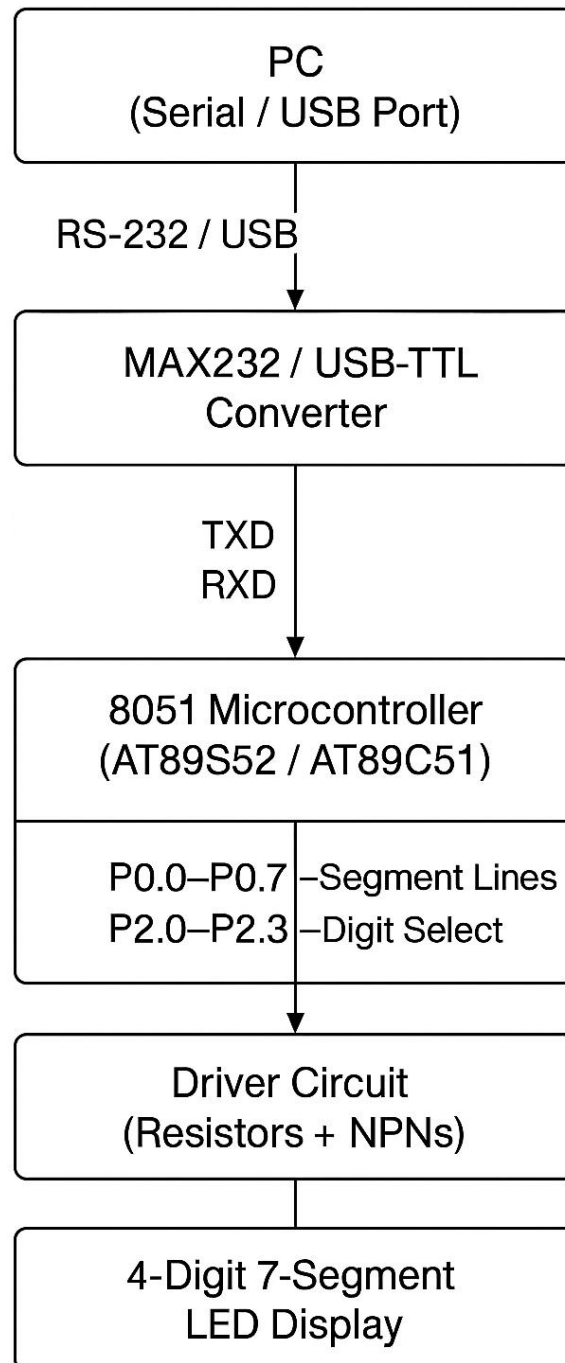
1.6 Significance

The significance of this project lies in its ability to integrate multiple concepts of embedded systems into a single working model. It provides exposure to:

1. Communication between computers and microcontrollers.
2. Conversion of digital data into human-readable output.
3. Efficient hardware utilization through multiplexing.

Such a system has potential applications in digital counters, small communication terminals, educational kits, and real-time information systems. Moreover, the skills gained through this project can be directly applied in industries dealing with embedded systems, consumer electronics, and automation.

CHAPTER 2 BLOCK DIAGRAM AND DISCRIPTION



2.1 BLOCK DIAGRAM

2.3 DESCRIPTION OF BLOCK DIAGRAM

2.3.1 PC (Serial / USB Port)

1. The PC is the source of the input data (characters).
2. You will type/send a character from the PC using a terminal software (like HyperTerminal, Tera Term, or Proteus Virtual Terminal).
3. Communication happens over either RS-232 or USB.

2.3.2 MAX232 / USB-TTL Converter

1. PCs normally communicate using RS-232 voltage levels ($\pm 12\text{V}$), which are not directly compatible with the microcontroller's TTL logic ($0\text{--}5\text{V}$).
2. The MAX232 IC is used to convert RS-232 signals into TTL-compatible signals.
3. If your PC has only USB, then a USB-to-TTL converter (like FT232 or CP2102) is used instead of MAX232.
4. This ensures reliable data transfer between PC and the 8051.

2.3.3 8051 Microcontroller (AT89S52 / AT89C51)

The 8051 MCU is the core of the system.

Its tasks include:

1. Receiving the character via RXD (P3.0).
2. Processing the ASCII value of the character.
3. Generating the next three consecutive characters.
4. Converting them into seven-segment display codes.
5. Sending those codes to the 7-seg display through segment and digit select lines.

Important pin usage:

6. P0.0 – P0.7 → Segment lines (a–g + dp).
7. P2.0 – P2.3 → Digit select (Digit 1–4).

2.3.4 Driver Circuit (Resistors + NPNs)

1. Directly driving a 4-digit 7-segment display from the microcontroller is not recommended due to current limitations.
2. Each segment line is connected via a resistor (220Ω) to limit current.
3. Each digit's common pin is switched using an NPN transistor (like BC547).

4. The transistor acts as a switch: the MCU enables one digit at a time while quickly cycling (multiplexing) all 4 digits.
5. To human eyes, this rapid switching looks like all digits are glowing simultaneously.

2.3.5 4-Digit 7-Segment LED Display

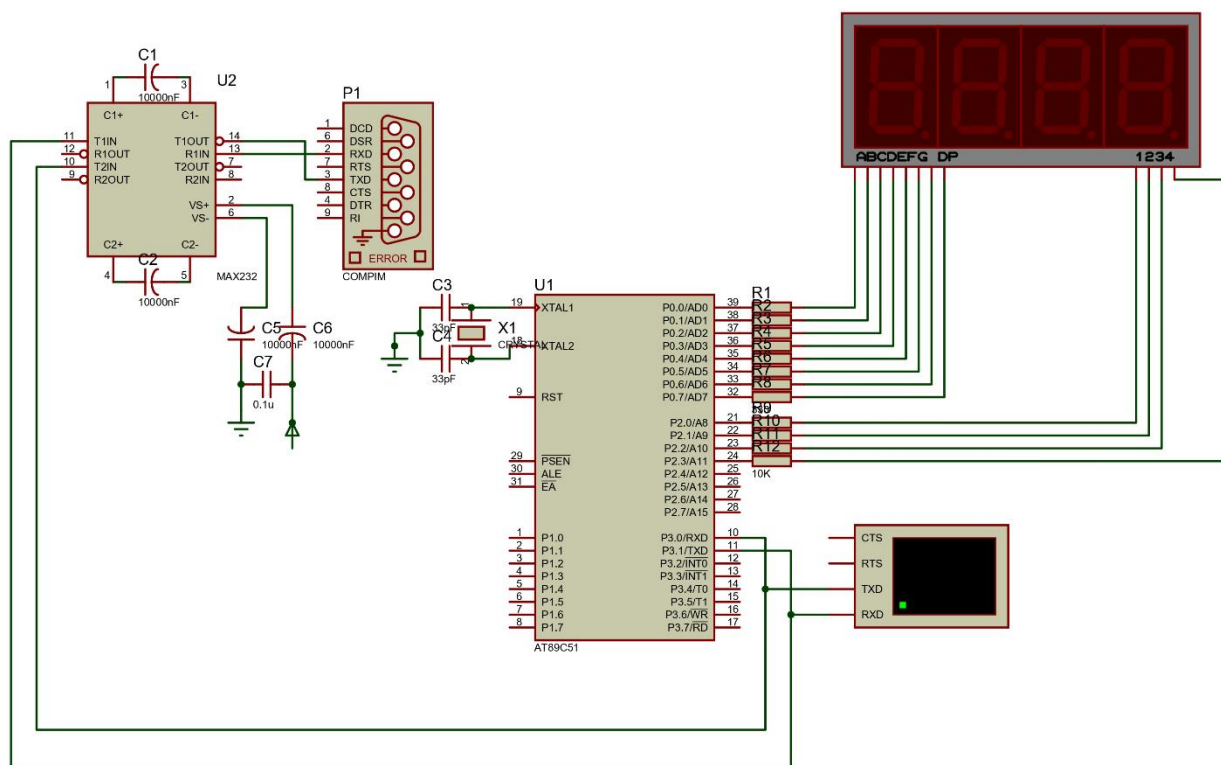
CHAPTER 1 The final output stages.

CHAPTER 2 The display shows the received character along with the next three characters.

- Example: Input '1' → Display 1234.
- Input 'B' → Display BCDE.

CHAPTER 3 Each digit is refreshed through multiplexing, driven by the microcontroller software.

CHAPTER 3 CIRCUIT DIAGRAM



3.1 CIRCUIT DIAGRAM

3.2 DESCRIPTION OF CIRCUIT DIAGRAM

The proposed system consists of a PC, a MAX232 level converter, an 8051 microcontroller (AT89C51), and a 4-digit seven-segment display. The PC is used to send serial data to the microcontroller. Since the RS-232 voltage levels of the PC are not compatible with TTL logic, the MAX232 IC is employed to convert the RS-232 levels into 5V TTL levels suitable for the 8051.

The microcontroller (U1) is the central unit of the system. It receives serial data through its RXD (P3.0) pin, processes the received character, generates the next three consecutive characters, and outputs the corresponding seven-segment codes. A crystal oscillator (11.0592 MHz) with capacitors C3 and C4 provides the clock frequency for accurate UART communication, while C5, C6, and C7 with the reset circuit ensure proper power-on reset and stable operation.

The seven-segment display is interfaced to Port 0 (P0.0–P0.7) of the microcontroller, which provides the segment data (a–g + dp). The digit enable lines are connected through Port 2 (P2.0–P2.3) to select each digit sequentially. Multiplexing is used, where the microcontroller rapidly switches between the four digits, making all appear active simultaneously to the human eye. Current-limiting resistors (R1–R4) are provided to protect the LEDs.

The serial port (P1) of the PC is connected through the MAX232 to ensure reliable bidirectional communication. A virtual terminal in Proteus is also used to emulate PC communication.

In operation, when a character (e.g., '1' or 'B') is sent from the PC, the microcontroller processes it and displays that character along with the next three consecutive characters (e.g., "1234" or "BCDE") on the 4-digit display. This demonstrates the combined application of serial communication, ASCII processing, and display multiplexing using the 8051 microcontrollers.

CHAPTER 4 IMAGE OF PROJECT

4.1 ACTUAL IMAGE OF PROJECT

CHAPTER 5 LIST OF COMPONENTS AND COMPONENTS DESCRIPTION

5.1 MICROCONTROLLER (AT89C51)

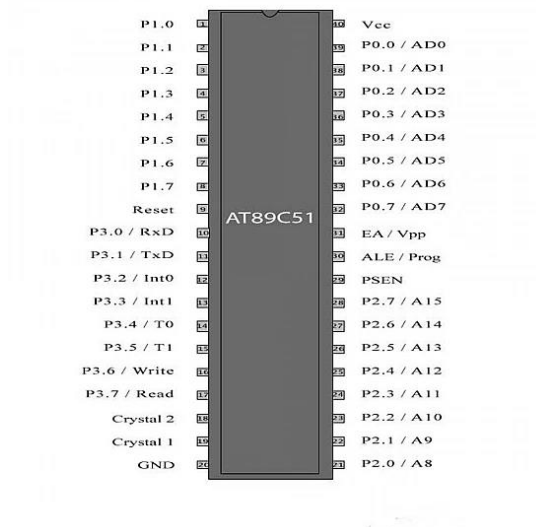


Figure 5.1 AT89C51

The AT89C51 is an 8-bit microcontroller belonging to the 8051 family, manufactured by Atmel. It has 4 KB Flash memory, 128 bytes of RAM, 32 I/O pins, two 16-bit timers, and a built-in full-duplex UART for serial communication. It acts as the brain of the project by receiving serial data from the PC, processing the ASCII values, and driving the 4-digit seven-segment display through multiplexing.

One of the biggest advantages of the AT89C51 is its versatility and wide availability. Being a member of the 8051 family, it supports standard instructions and can be easily programmed using common tools. Its UART interface makes it ideal for projects involving communication with computers, making it perfect for this display project.

5.2 4-DIGIT SEVEN SEGMENT LED DISPLAY

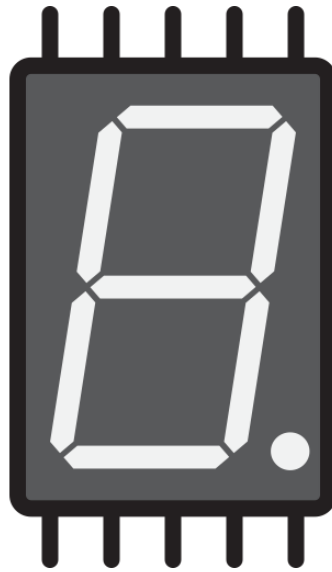


Figure 5.2 SEVEN SEGMENT LED DISPLAY

The seven-segment display works by lighting up different combinations of LED segments (labeled a–g) to form numbers and some characters. Each of the four digits shares common segment connections but has a separate common pin, which allows multiplexing by the microcontroller. This way, the microcontroller saves I/O pins and still manages to display four characters.

Multiplexing also reduces power consumption since only one digit is turned on at a time, but due to the fast switching speed, the human eye perceives it as a continuous display. This makes it a very efficient and low-cost way of displaying multiple characters in embedded systems.

5.3 MAX232 IC

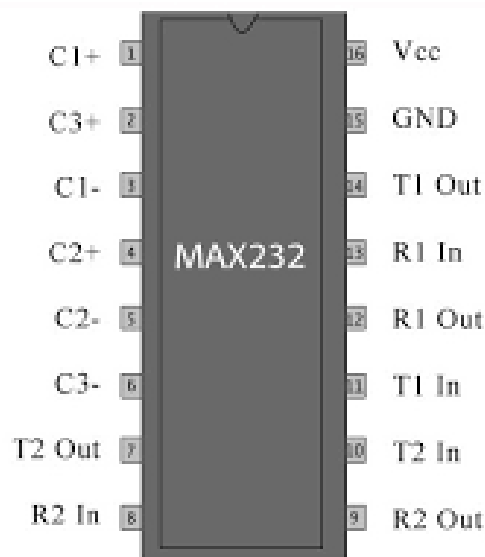


Figure 5.3 MAX232 IC

The MAX232 converts RS-232 signals ($\pm 12\text{V}$) from the PC into TTL-compatible signals (0–5V) that the microcontroller can safely process. Similarly, it converts TTL signals from the microcontroller into RS-232 levels so that the PC can interpret them correctly. Without this IC, direct communication between the PC and microcontroller could damage the circuit because of voltage mismatch.

The IC is highly reliable and widely used in embedded systems involving serial communication. It requires external capacitors for its internal charge pump circuitry, which generates the required voltage levels. Its dual-transceiver design makes it capable of handling two-way communication simultaneously.

6.4 Crystal Oscillator (11.0592 MHz)



Figure 6.4 Crystal Oscillator (11.0592 MHz)

The crystal oscillator is connected to the XTAL1 and XTAL2 pins of the AT89C51 microcontroller along with two 33 pF capacitors to ground. This provides the system clock that drives the internal operation of the microcontroller. The choice of 11.0592 MHz is significant, as it allows standard baud rates (e.g., 9600 bps, 4800 bps) to be generated with zero error.

Crystal oscillators are preferred over RC oscillators because they provide high frequency stability and accuracy. In communication-based projects like this one, accuracy in baud rate is crucial; otherwise, data corruption may occur during transmission or reception.

5.4 ALL OTHER COMPONENTS USED IN CIRCUIT\

- **Microcontroller**

- AT89C51 (8051 family, 40-pin DIP)

- **Display Unit**

- 4-Digit Common Cathode Seven-Segment Display

- **Level Converter IC**

- MAX232 (RS-232 to TTL converter)

- **Crystal Oscillator**

- 11.0592 MHz

- **Capacitors**

- $2 \times 33 \text{ pF}$ (for crystal oscillator)
- $4 \times 10 \text{ }\mu\text{F}$ electrolytic (for MAX232 charge pump)
- $1 \times 0.1 \text{ }\mu\text{F}$ ceramic (for bypass/filtering)
- $1 \times 10 \text{ }\mu\text{F}$ electrolytic (reset circuit)

- **Resistors**

- $4 \times 330 \text{ }\Omega$ (for current limiting on 7-seg)
- $1 \times 10 \text{ k}\Omega$ (pull-up for reset circuit)

- **Reset Components**

- Push Button (for reset switch)

- **Connector**

- DB9 Connector (serial port interface to PC)

- **Miscellaneous**

- Connecting wires / PCB
- Power supply: +5V regulated (from 7805 regulator or external source)

CHAPTER 6 COST OF PROJECT

Sr.No.	Description of Item	Qty.	PRICE
1	AT89C51	1	₹90
3	PCB BOARD	1	₹180
4	7 SEGMENT DSIPLAY	4	₹50
5	IC MAX232	1	₹150
6	CRYSRTAL	1	₹25
7	40 PIN IC SOCKET	1	₹20
8	LM7805	1	₹30
9	ZENNER DIODE	1	₹20
10	OTHER COMPONENTS & MISCELLANEOUS	-	₹350
11	TOTAL	9	₹915

Table No. 6.1

CHAPTER 7 CONCLUSION

Summary of the Project

In this project, a 4-digit 7-segment LED display was successfully interfaced with an 8051 microcontroller to receive serial data from a PC and display the received character along with the next three consecutive characters. For example, when the character '1' is received, the display shows '1234', and when 'B' is received, it shows 'BCDE'. The project demonstrates the integration of serial communication, real-time data processing, and multi-digit display control, which are essential concepts in embedded system design.

Technical Achievements

The system efficiently receives ASCII characters via UART, stores and processes the data, and updates the display dynamically. The implementation involved incrementing ASCII values, controlling display multiplexing, and maintaining proper timing to prevent flickering. Multiplexing techniques were employed to drive all four digits using a limited number of microcontroller pins, leveraging the principle of persistence of vision to provide a stable visual output.

Additionally, the project reinforced an understanding of serial communication and ASCII data handling. By receiving data from a PC, the microcontroller demonstrates real-world interfacing capabilities where embedded systems interact with higher-level computing devices for monitoring, control, or data display.

Hardware and Software Integration

From a hardware perspective, careful attention was given to connections between the 8051 microcontroller and the 7-segment display, ensuring reliable operation and minimal power consumption. On the software side, programming the microcontroller involved precise timing, serial data reception, and display scanning logic. Troubleshooting challenges such as display flicker and correct ASCII incrementation provided practical experience in debugging embedded systems.

Learning Outcomes and Applications

This project not only fulfilled its objectives but also offered significant learning experiences in embedded systems, digital electronics, and microcontroller programming. It serves as a foundation for more advanced projects, such as interactive digital counters, electronic signage, or human-machine interfaces in automation systems. The skills acquired here—serial communication, multiplexed display control, and real-time data processing—are widely applicable in electronics, robotics, and industrial automation.

Final Remarks

In conclusion, the project successfully combines theory with practical implementation, providing a functional and visually demonstrable system. It highlights the importance of precise hardware-software integration and offers a clear understanding of how digital data can be effectively represented and manipulated in real-time. Overall, this project strengthens core competencies in embedded system design and provides valuable experience for future electronic and automation applications.

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8.2 QR FOR REFERENCE



Figure 8.2 CODES